World-model MPC with a VLM scorer

MiniGrid-Empty-6x6-v0 - measured experimental report

Experiment summary

I train a compact RSSM world model on RGB MiniGrid rollouts and use it for model-predictive control. The main variant adds a frozen CLIP scorer to evaluate decoded future frames generated by the learned model.

The VLM score is computed on imagined future frames from the rollout, not only on the current real observation.

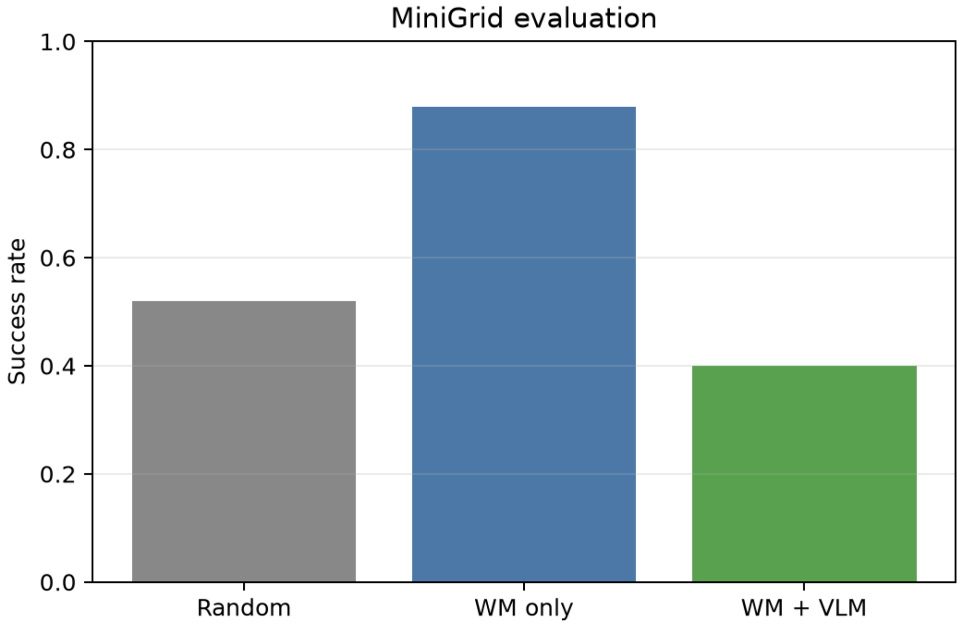
Environment	MiniGrid-Empty-6x6-v0, RGB observations resized to 64 x 64
Dataset	240 collected episodes; 153 successful trajectories
World model	CNN encoder, GRU deterministic state, Gaussian latent state, RGB decoder, reward and continuation heads
Planner	Random-shooting MPC, 32 candidates, horizon H=12
VLM scorer	openai/clip-vit-base-patch32; future steps 3, 6, 9, 12
Evaluation	25 episodes per method; seeds [0, 1, 2, 3, 4], five episodes per seed

Compared policies

Policy	Objective
random	Uniform random actions.
wm	Random-shooting MPC in the learned RSSM, using predicted discounted reward.
wm_vlm	The same planner plus the maximum CLIP goal score over decoded imagined frames.

Quantitative results

Method	Episodes	Success rate	Mean return	Mean length
Random	25	0.52	0.209	82.56
WM planning	25	0.88	0.520	52.00
WM + VLM	25	0.40	0.166	86.04



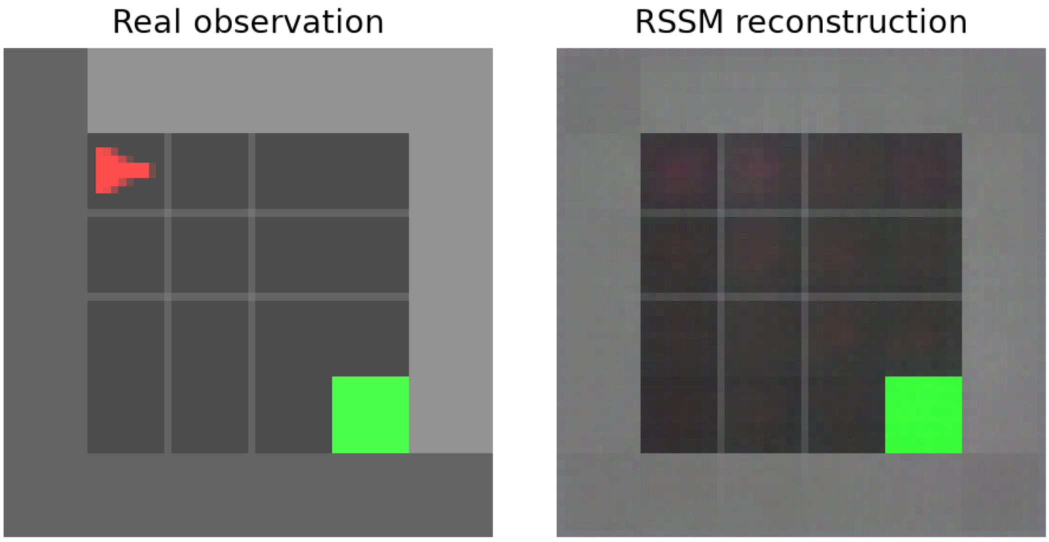
Reward-only world-model planning is strongest in this run: 0.88 success rate versus 0.52 for random and 0.40 for WM+VLM. The VLM path is implemented, but it is not a performance gain here. The result suggests that CLIP scores on symbolic MiniGrid reconstructions are noisy and can conflict with the learned reward objective.

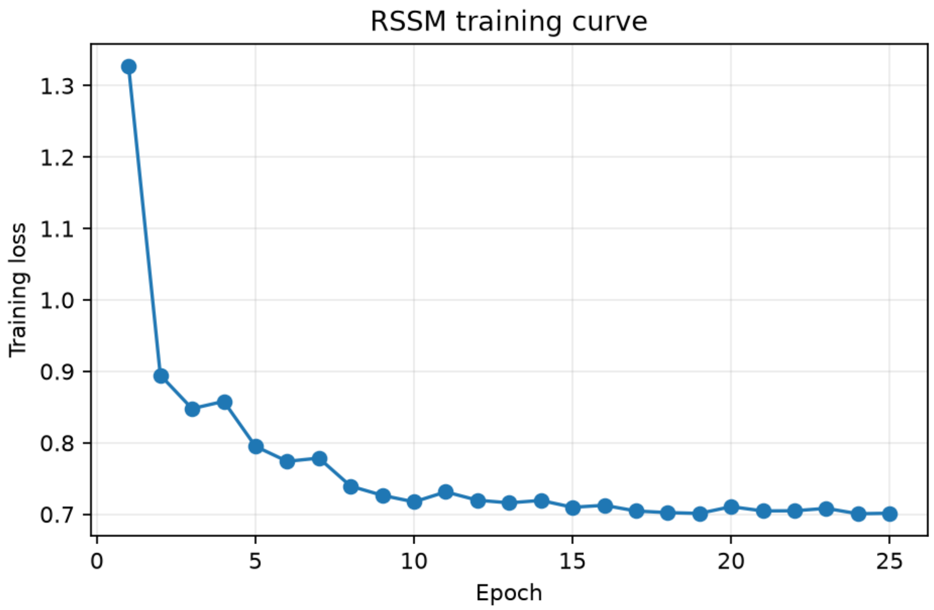
Aggregate values are stored in outputs/metrics.csv. Per-episode seeds, returns and lengths are stored in outputs/episodes.csv.

World model training and reconstruction check

The reconstruction check matters because the VLM scores decoded model predictions. If the decoder does not preserve objects, the semantic score becomes unreliable.

In this run the RSSM reconstructs the room layout and the green goal clearly. The red agent is less stable, which helps explain why CLIP scoring is noisy.



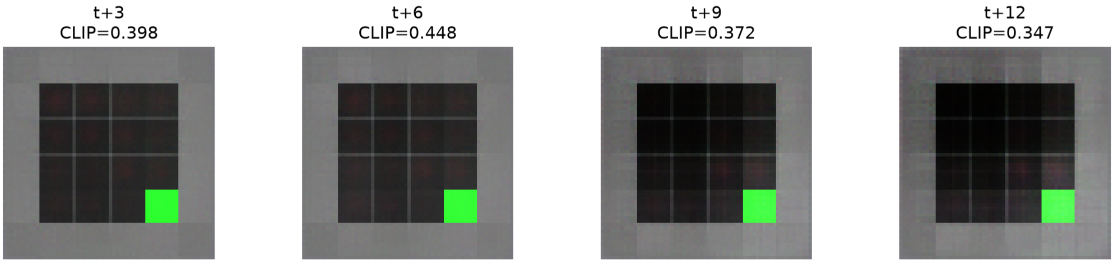


Training loss decreased from 1.327 to 0.701 over 25 epochs.

Imagined rollouts and limitations

The decoded future frames below come from a learned rollout and are scored by CLIP. The green goal remains visible, while the agent signal is weak. This makes it plausible that CLIP rewards the presence of the green square more than the exact relation "agent standing on goal".

Decoded imagined future frames scored by the VLM



Observed failure modes

- CLIP can reward green pixels instead of the spatial relation agent-on-goal.

- RSSM rollout errors accumulate; the red agent becomes blurry or disappears.

- Decoder artefacts directly affect the VLM objective.

- Random shooting with 32 candidates is brittle when the objective is noisy.

Future work

- Replace random shooting with CEM.

- Fine-tune a small contrastive scorer on MiniGrid relation labels.

- Add uncertainty penalties to imagined states.

- Compare with a symbolic oracle scorer for diagnosis only.

- Try an object-centric world model or a full Dreamer actor-critic agent.

Reproducibility

Measured run command: python run_pipeline.py --methods random wm wm_vlm

The repository contains the source code, configuration, measured CSV files, plots, screenshots, GIF visualization, PDF report and full run log. Large regenerable files such as the collected dataset and RSSM checkpoint are not committed; they are produced again by running the pipeline.